B9339 Quantitative Hedge Fund Strategies

Homework 4

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1. Inverting returns

Notation. Let X_t be the price of one unit of currency A in units of currency B (for NZD/JPY: JPY per NZD). The simple return in this direction is $R = X_{t+1}/X_t - 1$. The opposite direction has price $Y_t = 1/X_t$ (NZD per JPY) and return R'.

(a) Why the return is not inverted

$$R' = \frac{Y_{t+1}}{Y_t} - 1 = \frac{X_t}{X_{t+1}} - 1 = \frac{1}{1+R} - 1 = -\frac{R}{1+R} = -R + R^2 - R^3 + \cdots$$

Prices invert exactly ($Y = 1/X$), but returns do not: R' is neither $1/R$ nor $-R$. Each return is measured relative to its own starting value, in its own numeraire. When A appreciates by 10% against B, B depreciates by only $1 - 1/1.1 = 9.09091\%$ against A, because the 10% is a fraction of the (smaller) starting price of A while the 9.09% is a fraction of the (larger) starting price of B. The function $g(R) = 1/(1+R) - 1$ is convex, so

$$R + R' = \frac{R^2}{1+R} \geq 0 \quad \Rightarrow \quad E[R] + E[R'] = E\left[\frac{R^2}{1+R}\right] > 0$$

Both sides of the pair can therefore have a positive expected return at the same time (Jensen's inequality, the 'Siegel paradox'). This is why carry-trade returns must be computed in the direction actually held and in the investor's numeraire. Only log returns are antisymmetric: $\ln(1+R') = -\ln(1+R)$.

(b) Approximate moments of the inverse return

Only $\mu = E[R]$ and $\sigma^2 = \mathrm{Var}[R]$ are known, so expand $g(R)$ to second order around μ (the delta method). The derivatives are $g'(R) = -(1+R)^{-2}$, $g''(R) = 2(1+R)^{-3}$ and $g'''(R) = -6(1+R)^{-4}$:

$$g(R) \approx g(\mu) + g'(\mu)(R-\mu) + \frac{1}{2}g''(\mu)(R-\mu)^2$$

Taking expectations (using $E[R-\mu] = 0$ and $E[(R-\mu)^2] = \sigma^2$) and variances (keeping the linear term) gives:

$$E[R'] \approx -\frac{\mu}{1+\mu} + \frac{\sigma^2}{(1+\mu)^3} \approx -\mu + \mu^2 + \sigma^2, \qquad \mathrm{Var}[R'] \approx \frac{\sigma^2}{(1+\mu)^4} \approx \sigma^2$$

The small-μ forms follow from $1/(1+\mu) \approx 1-\mu$. The key point is that the expected inverse return is not $-\mu$ but $-\mu$ plus a variance (convexity) term. For daily FX returns, σ^2 is of the same order of magnitude as μ, so this term is not negligible; volatility is essentially unchanged by inversion. For the empirical check I also report the next terms. They need the third and fourth central moments m_3 and m_4, which (b) does not assume are known:

$$E[R'] \approx -\frac{\mu}{1+\mu} + \frac{\sigma^2}{(1+\mu)^3} - \frac{m_3}{(1+\mu)^4} + \frac{m_4}{(1+\mu)^5}$$

$$\mathrm{Var}[R'] \approx \frac{\sigma^2}{(1+\mu)^4} - \frac{2m_3}{(1+\mu)^5} + \frac{3m_4 - \sigma^4}{(1+\mu)^6}$$

(c) Empirical check: NZD/JPY and JPY/NZD

The workbook (hw3.data.carry.FX.dly.xlsx, the HW3 file) quotes NZD as USD per NZD and JPY as JPY per USD. The cross rates are therefore

$$X_t^{NZD/JPY} = S_t^{NZD}\,[\mathrm{USD/NZD}] \times S_t^{JPY}\,[\mathrm{JPY/USD}] \;\; (\text{JPY per NZD}), \qquad X_t^{JPY/NZD} = 1/X_t^{NZD/JPY} \;\; (\text{NZD per JPY})$$

with daily returns $R_t = X_t/X_{t-1} - 1$ for both series (1999-01-04 to 2010-06-04, 2980 returns; spot from 60.13984 to 61.64404 JPY per NZD). Moments are population moments (the mean of $(R-\mu)^k$), which is what the Taylor expansion of the sample mean of $g(R)$ uses. The JPY/NZD series computed directly from $1/X$ matches $1/(1+R) - 1$ to 10^{-14}.

Table 1: Sample moments of the daily returns in both directions.

Series	mean (daily)	variance (daily)	std (daily)	skewness	kurtosis	mean × 260	std × √260
NZD/JPY (JPY per NZD)	6.94248e-05	1.21800e-04	0.01104	-0.44977	10.52510	0.01805	0.17796
JPY/NZD (NZD per JPY)	5.31169e-05	1.23435e-04	0.01111	0.75754	10.89289	0.01381	0.17915

Both directions have a *positive* average return. Their sum is 1.22542×10^{-4}, almost exactly σ^2 = 1.21800×10^{-4}, while the mean log return is only 8.28993×10^{-6} per day. This is the convexity effect from (a).

Table 2: Moments of the inverse series predicted from the moments of the other direction. Predicted = formula from (b) evaluated at the sample μ and σ^2 (and m_3, m_4 in the last rows). Error = Predicted – Actual.

Predict	Moment	Approximation (mean / variance)	Actual	Predicted	Error	Rel. error
NZD/JPY → JPY/NZD	$E[R']$	$-\mu\ /\ \sigma^2$	5.31169e-05	-6.94248e-05	-1.22542e-04	-230.70194%
NZD/JPY → JPY/NZD	$E[R']$	$-\mu/(1+\mu)$	5.31169e-05	-6.94200e-05	-1.22537e-04	-230.69287%
NZD/JPY → JPY/NZD	$E[R']$	(b) second order	5.31169e-05	5.23549e-05	-7.61949e-07	-1.43448%
NZD/JPY → JPY/NZD	$E[R']$	(b) $-\mu+\mu^2+\sigma^2$	5.31169e-05	5.23803e-05	-7.36585e-07	-1.38672%
NZD/JPY → JPY/NZD	$E[R']$	with m_3, m_4 terms	5.31169e-05	5.31154e-05	-1.43419e-09	-0.00270%
NZD/JPY → JPY/NZD	$\mathrm{Var}[R']$	$-\mu\ /\ \sigma^2$	1.23435e-04	1.21800e-04	-1.63468e-06	-1.32433%
NZD/JPY → JPY/NZD	$\mathrm{Var}[R']$	(b) second order	1.23435e-04	1.21766e-04	-1.66850e-06	-1.35172%
NZD/JPY → JPY/NZD	$\mathrm{Var}[R']$	with m_3, m_4 terms	1.23435e-04	1.23429e-04	-6.32531e-09	-0.00512%
JPY/NZD → NZD/JPY	$E[R']$	$-\mu\ /\ \sigma^2$	6.94248e-05	-5.31169e-05	-1.22542e-04	-176.50996%
JPY/NZD → NZD/JPY	$E[R']$	$-\mu/(1+\mu)$	6.94248e-05	-5.31141e-05	-1.22539e-04	-176.50589%
JPY/NZD → NZD/JPY	$E[R']$	(b) second order	6.94248e-05	7.03012e-05	8.76430e-07	1.26242%
JPY/NZD → NZD/JPY	$E[R']$	(b) $-\mu+\mu^2+\sigma^2$	6.94248e-05	7.03209e-05	8.96098e-07	1.29075%
JPY/NZD → NZD/JPY	$E[R']$	with m_3, m_4 terms	6.94248e-05	6.94285e-05	3.69330e-09	0.00532%
JPY/NZD → NZD/JPY	$\mathrm{Var}[R']$	$-\mu\ /\ \sigma^2$	1.21800e-04	1.23435e-04	1.63468e-06	1.34210%
JPY/NZD → NZD/JPY	$\mathrm{Var}[R']$	(b) second order	1.21800e-04	1.23409e-04	1.60846e-06	1.32057%
JPY/NZD → NZD/JPY	$\mathrm{Var}[R']$	with m_3, m_4 terms	1.21800e-04	1.21814e-04	1.37589e-08	0.01130%

Do the data agree? Yes. For the mean, the naive guess $-\mu$ has the wrong sign (error -1.22542×10^{-4}, about 3.18608% per year). The second-order formula from (b) cuts the error to -7.61949×10^{-7}, removing

99.37821% of the naive error, and the small-μ version is just as good. The remaining relative gap of about 1.4% comes from the third and fourth moments: daily NZD/JPY returns have negative skewness (-0.44977, the carry-crash signature) and kurtosis above 10, so the higher-order terms are not negligible. Adding them reduces the error to about 10^{-9}. For the variance, (b) predicts $\mathrm{Var}[R'] \approx \sigma^2$, which is right to about 1.3%. The factor $(1+\mu)^{-4}$ is practically 1, and the residual is the $-2m_3$ skewness term (inversion flips the sign of skewness: -0.44977 vs 0.75754). Figure 1 repeats the comparison on rolling one-year windows: the largest error of the second-order formula is 5.43008×10^{-6}, vs 4.99609×10^{-4} for $-\mu$.

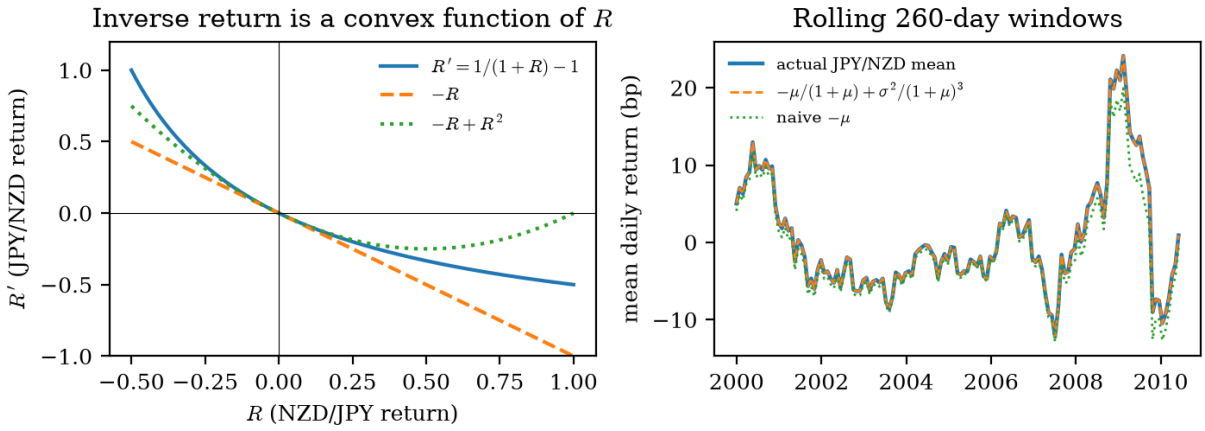


Figure 1: Left, the inverse return as a function of R, with the linear ($-R$) and quadratic approximations. Right, mean daily JPY/NZD return over rolling 260-day windows (every 20 days): actual vs the second-order prediction and $-\mu$, both computed from the NZD/JPY moments in the same window.

2. The Rule of 72

With discrete compounding at rate r per period, doubling requires $(1+r)^n = 2$, so the exact answer is

$$n^*(r) = \frac{\ln 2}{\ln(1+r)}$$

First-order approximations. Using $\ln(1+r) \approx r$ gives $n \approx \ln 2/r = 69.3147/(100r)$ ('Rule of 69.3'), or, with $\ln 2 \approx 0.69$, $n \approx 69/(100r)$. Since $\ln(1+r) < r$, both are always too short.

My two approximations (second order). The series $1/\ln(1+r) = 1/r + 1/2 - r/12 + r^2/24 - \cdots$ (equivalently $\ln(1+r) \approx r - r^2/2$ and $1/(1-r/2) \approx 1 + r/2$) gives:

$$\text{(iv)}\quad n \approx \ln 2\left(\frac{1}{r} + \frac{1}{2}\right) = \frac{69.3147 + 34.6574\,r}{100\,r}, \qquad \text{(v)}\quad n \approx 0.69\left(\frac{1}{r} + \frac{1}{2}\right) = \frac{69 + 34.5\,r}{100\,r}$$

Approximation (iv) is the better one: its error is about $\ln 2 \cdot r/12 > 0$, under 0.005 periods for $r \le 9\%$. Approximation (v) loses $0.00315(1/r + 1/2)$ periods from rounding ln 2 down to 0.69, so it is worse at low rates. This rounding loss happens to offset the positive $r/12$ error, cancelling exactly at r = 26.37188%, so (v) is slightly more accurate above roughly 18%. Form (iv) also explains the Rule of 72: it is a 'rule of N' with $N(r) = 69.3147 + 34.6574r$, which equals 72 when $r \approx 7.75\%$. The number 72 is the constant that is right for typical interest rates (and it is divisible by 2, 3, 4, 6, 8, 9 and 12).

(a) Sensitivity to the interest rate

I evaluated all five approximations and the exact n^* on a grid of r from 0.1% to 100% (Python, numpy). Table 3 shows the error $n_{\text{approx}} - n^*$ in periods, and Figure 2 plots the absolute and relative errors.

Table 3: Doubling periods and approximation errors ($n_{\text{approx}} - n^*$, in periods). The row r = 7.84687% is where the Rule of 72 is exact.

r	n^*	$72/(100r)$	error $72/(100r)$	error $\ln 2/r$	error $0.69/r$	error $\ln 2(1/r + 1/2)$	error $0.69(1/r + 1/2)$
0.50000%	138.97572	144.00000	5.02428	-0.34629	-0.97572	0.00029	-0.63072
1.00000%	69.66072	72.00000	2.33928	-0.34600	-0.66072	0.00057	-0.31572
2.00000%	35.00279	36.00000	0.99721	-0.34543	-0.50279	0.00114	-0.15779
3.00000%	23.44977	24.00000	0.55023	-0.34487	-0.44977	0.00171	-0.10477
4.00000%	17.67299	18.00000	0.32701	-0.34431	-0.42299	0.00227	-0.07799
5.00000%	14.20670	14.40000	0.19330	-0.34376	-0.40670	0.00282	-0.06170
6.00000%	11.89566	12.00000	0.10434	-0.34321	-0.39566	0.00337	-0.05066
7.00000%	10.24477	10.28571	0.04095	-0.34267	-0.38763	0.00391	-0.04263
7.84687%	9.17563	9.17563	0.00000	-0.34221	-0.38232	0.00436	-0.03732
8.00000%	9.00647	9.00000	-0.00647	-0.34213	-0.38147	0.00445	-0.03647
9.00000%	8.04323	8.00000	-0.04323	-0.34160	-0.37657	0.00498	-0.03157
10.00000%	7.27254	7.20000	-0.07254	-0.34107	-0.37254	0.00550	-0.02754
12.00000%	6.11626	6.00000	-0.11626	-0.34003	-0.36626	0.00654	-0.02126
15.00000%	4.95948	4.80000	-0.15948	-0.33850	-0.35948	0.00807	-0.01448
20.00000%	3.80178	3.60000	-0.20178	-0.33605	-0.35178	0.01053	-0.00678
25.00000%	3.10628	2.88000	-0.22628	-0.33369	-0.34628	0.01288	-0.00128
30.00000%	2.64193	2.40000	-0.24193	-0.33144	-0.34193	0.01514	0.00307
40.00000%	2.06004	1.80000	-0.26004	-0.32717	-0.33504	0.01940	0.00996
50.00000%	1.70951	1.44000	-0.26951	-0.32322	-0.32951	0.02336	0.01549
75.00000%	1.23861	0.96000	-0.27861	-0.31442	-0.31861	0.03216	0.02639
100.00000%	1.00000	0.72000	-0.28000	-0.30685	-0.31000	0.03972	0.03500

Table 4: Relative error $n_{\text{approx}}/n^* - 1$. The search grid starts at r = 0.1%.

Approximation	rel. error at r = 1%	rel. error at r = 10%	rel. error at r = 25%	range of r with |rel. error| ≤ 1%
(i) $72/(100r)$	3.35811%	-0.99746%	-7.28471%	5.74541% to 10.00518%
(ii) $\ln 2/r$	-0.49669%	-4.68982%	-10.74258%	0.10000% to 2.02609%
(iii) $0.69/r$	-0.94848%	-5.12257%	-11.14785%	0.10000% to 1.10401%
(iv) $\ln 2(1/r + 1/2)$	0.00083%	0.07569%	0.41460%	0.10000% to 41.44703%
(v) $0.69(1/r + 1/2)$	-0.45322%	-0.37870%	-0.04133%	0.10000% to 52.08548%

Findings. The first-order rules (ii) and (iii) have an almost constant absolute error of about –0.34 periods ($-\ln 2/2 = -0.34657$ in the limit), so they are fine only when n is large (r below about 2%), and their relative error grows roughly like $-r/2$. The Rule of 72 overestimates for r < 7.84687% (by 2.33928 periods at 1%) and underestimates above it; it is within 1% only for r between 5.74541% and 10.00518%. The second-order rules (iv) and (v) are within 1% for every rate up to 41% and 52% respectively, so they are much less sensitive to the rate. The rule-of-N constant needed for an exact answer rises with r: 1%: 69.66072, 2%: 70.00558, 5%: 71.03350, 8%: 72.05175, 10%: 72.72541, 15%: 74.39227, 20%: 76.03568.

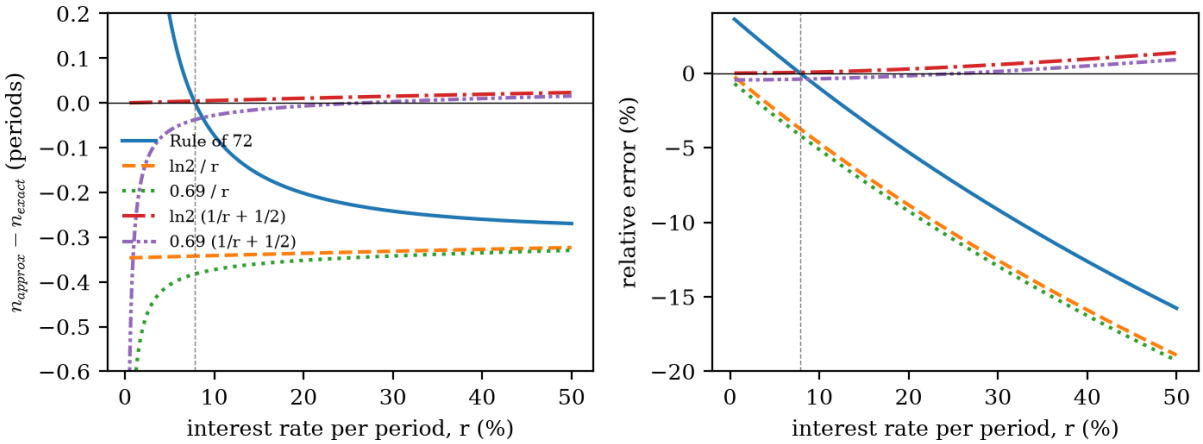


Figure 2: Absolute (left, periods) and relative (right) errors of the five approximations; the dashed vertical line marks the rate at which the Rule of 72 is exact.

(b) Rate at which the Rule of 72 is exact

Solve $72/(100r) = \ln 2/\ln(1+r)$, i.e. $h(r) = 0.72\ln(1+r) - r\ln 2 = 0$, with Brent's method (tolerance 10^{-15}). Since $h(0) = 0$, $h'(0) = 0.72 - \ln 2 > 0$ and h is strictly concave, there is exactly one positive root:

$$r^* = 0.0784687 \quad (7.84687\%), \qquad n^* = 9.17563 \text{ periods}$$

At this rate the Rule of 72 matches the true doubling time. It is closest to the truth for rates near 8%, which is why practitioners chose 72.

3. FX carry: Jurek (2014), crash-neutral carry

Setup (Jurek's notation). The trader is long the high-yield currency J, funded in I. S_t^{ji} is the price of J in units of I, and y^j, y^i are the interest rates; below, S_t and $S_{t+\tau}$ stand for S^{ji}. Per unit of capital the trader holds $(1-q_p\delta_p)$ units of J exposure: 1 unit for the carry trade plus $-q_p\delta_p > 0$ units to hedge the puts' (negative) delta. They buy q_p puts with strike K_p at price $P_t(K_p,\tau)$, and borrow everything in I. Equation (2) is:

$$\tilde{r}_{t+\tau} = e^{y^j\tau}(1-q_p\delta_p)\frac{S_{t+\tau}}{S_t} + q_p\max\left(\frac{K_p}{S_t} - \frac{S_{t+\tau}}{S_t}, 0\right) - e^{y^i\tau}\left((1-q_p\delta_p) + q_p\frac{P_t(K_p,\tau)}{S_t}\right)$$

(a) The expression for q_p in Equation (3)

The goal is a position with *no* exposure to $S_{t+\tau}$ below the strike. For $S_{t+\tau} < K_p$ the put is in the money, and the terms of (2) that depend on $S_{t+\tau}$ are

$$\left[e^{y^j\tau}(1-q_p\delta_p) - q_p\right]\frac{S_{t+\tau}}{S_t}$$

The first part is the long J holding, which has grown at the J interest rate to $e^{y^j\tau}(1-q_p\delta_p)$ units. The second part is the q_p puts, each gaining one unit per unit fall in S. Setting the bracket to zero (one put per unit of terminal J holding) and solving gives (3):

$$q_p = e^{y^j\tau}(1-q_p\delta_p) \;\Rightarrow\; q_p\left(1+e^{y^j\tau}\delta_p\right) = e^{y^j\tau} \;\Rightarrow\; q_p = \frac{e^{y^j\tau}}{1+e^{y^j\tau}\delta_p}$$

The equation is circular. The puts have a negative delta, so $-q_p\delta_p$ extra units of J are bought to keep the initial exposure equal to one unit of carry: the total delta is $(1-q_p\delta_p)+q_p\delta_p = 1$. But those extra J units also need put protection, which needs more delta hedging, and so on. Solving the fixed point sums this geometric series:

$$q_p = e^{y^j\tau}\sum_{k=0}^{\infty}\left(-e^{y^j\tau}\delta_p\right)^k$$

The Garman–Kohlhagen put delta is $\delta_p = -e^{-y^j\tau}N(-d_1)$, so $-1 < e^{y^j\tau}\delta_p < 0$; hence the series converges and q_p is finite and larger than $e^{y^j\tau}$.

(b) Why Equation (4) shows the return is bounded from below

Step 1. By (3), $e^{y^j\tau}(1-q_p\delta_p) = q_p$, so the first two terms of (2) combine:

$$q_p\,\frac{S_{t+\tau} + \max(K_p - S_{t+\tau}, 0)}{S_t} = q_p\max\left(\frac{K_p}{S_t}, \frac{S_{t+\tau}}{S_t}\right)$$

which is the first line of (4). **Step 2.** FX put-call parity, $P_t = C_t - S_t e^{-y^j\tau} + K_p e^{-y^i\tau}$, together with $1-q_p\delta_p = q_p e^{-y^j\tau}$, turns the funding cost into

$$e^{y^i\tau}\left(q_p e^{-y^j\tau} + q_p\frac{C_t - S_t e^{-y^j\tau} + K_p e^{-y^i\tau}}{S_t}\right) = q_p\frac{K_p}{S_t} + q_p\,e^{y^i\tau}\frac{C_t(K_p,\tau)}{S_t}$$

Step 3. Subtracting gives the second line of (4):

$$\tilde{r}_{t+\tau} = q_p \max\left(\frac{S_{t+\tau} - K_p}{S_t}, 0\right) - q_p\, e^{y^i \tau}\frac{C_t(K_p,\tau)}{S_t} \;\geq\; -\,q_p\, e^{y^i \tau}\frac{C_t(K_p,\tau)}{S_t}$$

Economically, the crash-neutral trade is just q_p call options on J, bought with borrowed I. The first term is an option payoff, so it can never be negative. The second term is fixed at time t: the call premium plus interest on the borrowing. So the worst-case return is known when the trade is opened, no matter how far J crashes. Unhedged carry, $e^{y^j \tau} S_{t+\tau}/S_t - e^{y^i \tau}$, has no such floor. Numerical check (Figure 3: one-month options, y^j = 4.5%, y^i = 0.5%, 12% volatility, $K_p/S_t = 0.97$, Garman–Kohlhagen prices): $\delta_p = -0.21095$, $q_p = 1.27339$, floor = -0.03933. Equations (2) and (4) agree to 3.53884×10^{-16} at every terminal rate.

(c) Why the payoff is steeper than unhedged carry above the strike

For $S_{t+\tau} > K_p$, (4) is linear in $S_{t+\tau}/S_t$ with slope q_p. The slope of unhedged carry is $e^{y^j \tau}$. The ratio is

$$\frac{q_p}{e^{y^j \tau}} = \frac{1}{1 + e^{y^j \tau}\delta_p} > 1 \quad \text{since} \quad \delta_p < 0$$

which is Jurek's 'magnified by a factor $q_p/\exp(y^j\tau)$' (1.26863 in the example). The reason is the delta hedge. The trader bought $-q_p\delta_p$ extra units of J to offset the puts' negative delta at inception. If J ends above the strike, the puts expire worthless, but the extra J units are still held and add upside. In effect, the hedge turns put-protected carry into levered call exposure. The cost is a lower floor than an unhedged put overlay (-0.03933 vs -0.03101 in the example), because more options are bought and the extra position must also be protected.

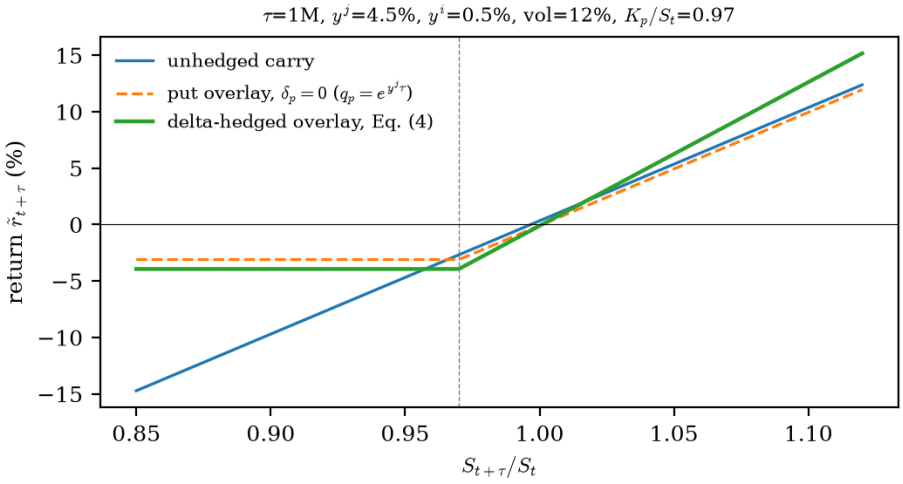


Figure 3: Terminal return vs $S_{t+\tau}/S_t$ for unhedged carry, carry plus an unhedged put overlay ($\delta_p = 0$), and the delta-hedged crash-neutral trade of Eq. (4). The dashed line marks the strike.

(d) Unhedged overlay: set $\delta_p = 0$

δ_p enters (2) only through the extra $-q_p\delta_p$ units of J bought to hedge the option delta. Not hedging the overlay means buying none, i.e. setting $\delta_p = 0$. Then (3) gives $q_p = e^{y^j \tau}$, exactly one put per unit of terminal J holding (principal plus interest), and (2) becomes a carry trade with a protective put:

$$\tilde{r}_{t+\tau} = e^{y^j \tau} \max\left(\frac{S_{t+\tau}}{S_t}, \frac{K_p}{S_t}\right) - e^{y^i \tau}\left(1 + e^{y^j \tau}\frac{P_t(K_p, \tau)}{S_t}\right)$$

This is still bounded below, but above the strike its slope is $e^{y^j \tau}$, the same as plain carry (magnification factor 1), and it sits below plain carry by the financed put premium. As the strike goes to 0, the put becomes worthless ($P_t \to 0$), its delta goes to 0 ($N(-d_1) \to 0$), and the put payoff is 0. The hedge therefore disappears automatically, $q_p \to e^{y^j \tau}$, and (2) reduces to the unhedged carry return $e^{y^j \tau} S_{t+\tau}/S_t - e^{y^i \tau}$. In other words, $\delta_p = 0$ with $K_p > 0$ is the put-protected trade, and $K_p \to 0$ recovers the original carry trade as a special case. As Jurek notes, if I is not the investor's home currency h, each return is converted by multiplying by $S^{ih}_{t+\tau}/S^{ih}_t$.

Reproducibility

Code: https://github.com/KimYoungMuri/carrytrade_returns. scripts/run_all.py runs all computations (tables in outputs/tables, figures in outputs/figures) and scripts/build_report.py builds this PDF. The data loader (src/fxcarry/data.py) is the verified HW3 loader.